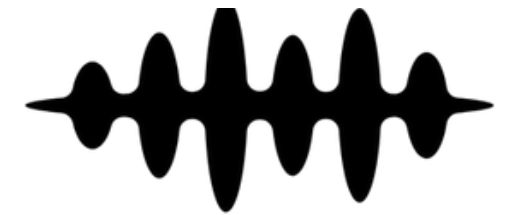




# VOICELESS TO VOICE



Real-Time Indian Sign Language (ISL) to  
Speech Conversion using Deep Learning

**Guide:** Dr. Brijbihari Dubey

**BTECH CSE (DS) 7TH**

BHANKALE YASH  
KARANDIKAR SAKSHAT

SHINGOTE AMRUTA  
MORE HIMANSHU

# THE SILENT BARRIER

Millions of individuals worldwide rely on sign language as their primary mode of communication, yet a massive gap exists between them and the hearing world.

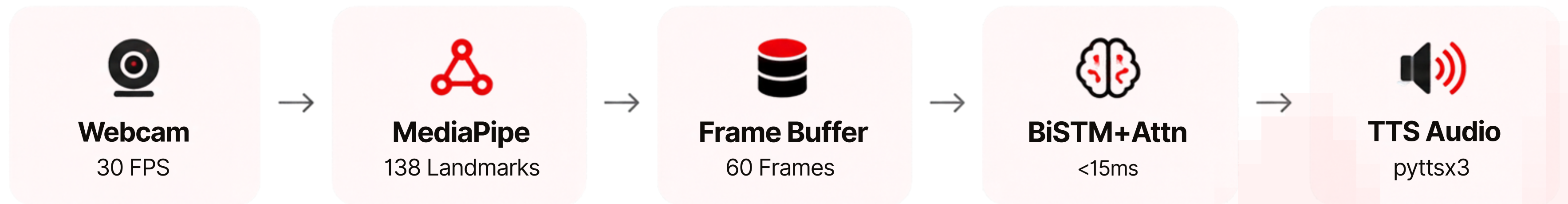
- **Scalability:** Human interpreters are vital but are often expensive and unavailable for spontaneous, daily interactions.
- **Hardware Constraints:** Many existing digital solutions require bulky sensors or high-end processing power that isn't accessible to everyone.
- **The Real-Time Gap:** Current technology often solves the "recognition" problem but fails to provide the human element: instant, spoken feedback.



## SYSTEM OVERVIEW

# How Does **Voiceless to Voice** Work?

End-to-end pipeline: gesture to spoken word in under 500ms

**30 FPS**

Live Inference

**<15ms**

Per-Frame Latency

**35**

ISL Classes

**100%**

Offline Capable

# Full System Architecture

A complete pipeline from training to real-time inference with a robust fallback strategy.

## 01 TRAINING PHASE (Google Colab GPU)



Dataset collection → 60-frame .npY sequences



Upload to Google Drive (T4 GPU training)



BILSTM+Attention → **unified\_sign\_model.pt**



Download model (~3MB) for local use

## 02 INFERENCE PHASE (Local Machine)



Webcam → MediaPipe Holistic



Rolling deque(60 frames)



PyTorch Model Inference



Majority voting + Cooldown



pyttsx3 TTS → Speaker

# Dataset Collection & Feature Engineering



## MediaPipe Holistic Landmarks

Left Hand (21 keypoints × 3) **63**



Right Hand (21 keypoints × 3) **63**



Upper Pose (4 keypoints × 3) **12**



**Total: 138 features / frame**

Sequence: 60 frames × 138 = 8,280 values/gesture



## Dataset Specs



**35**

ISL Classes



**30+**

Sequences/Class



**60**

Frames/Sequence

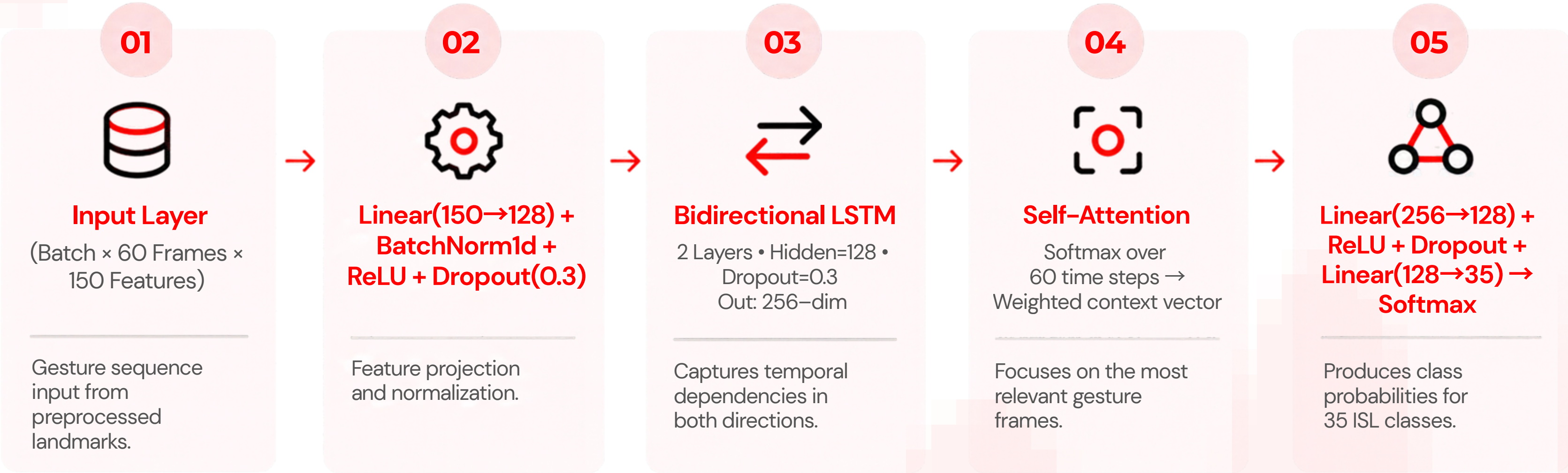


**~2s**

Window Duration

# BiLSTM + Self-Attention Neural Network

From gesture sequences to class probabilities using a deep neural network.



**3000**  
ISL Classes

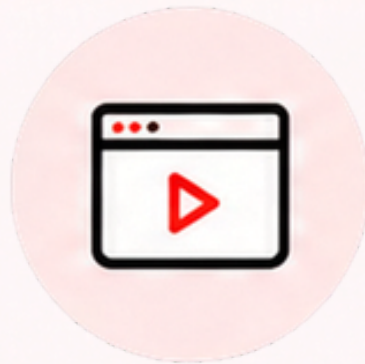
A deep learning pipeline designed for accurate and real-time gesture recognition.



**From Gestures to Meaning**  
Powered by Deep Learning

# Gradio Web Dashboard & ISL Vocabulary

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## Tab 1 — Video Translation

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Upload ISL video → Annotated output with landmark tracking & recognized sentence text



## Tab 2 — Supported Vocab

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Full list of trained ISL sign classes displayed as interactive labels

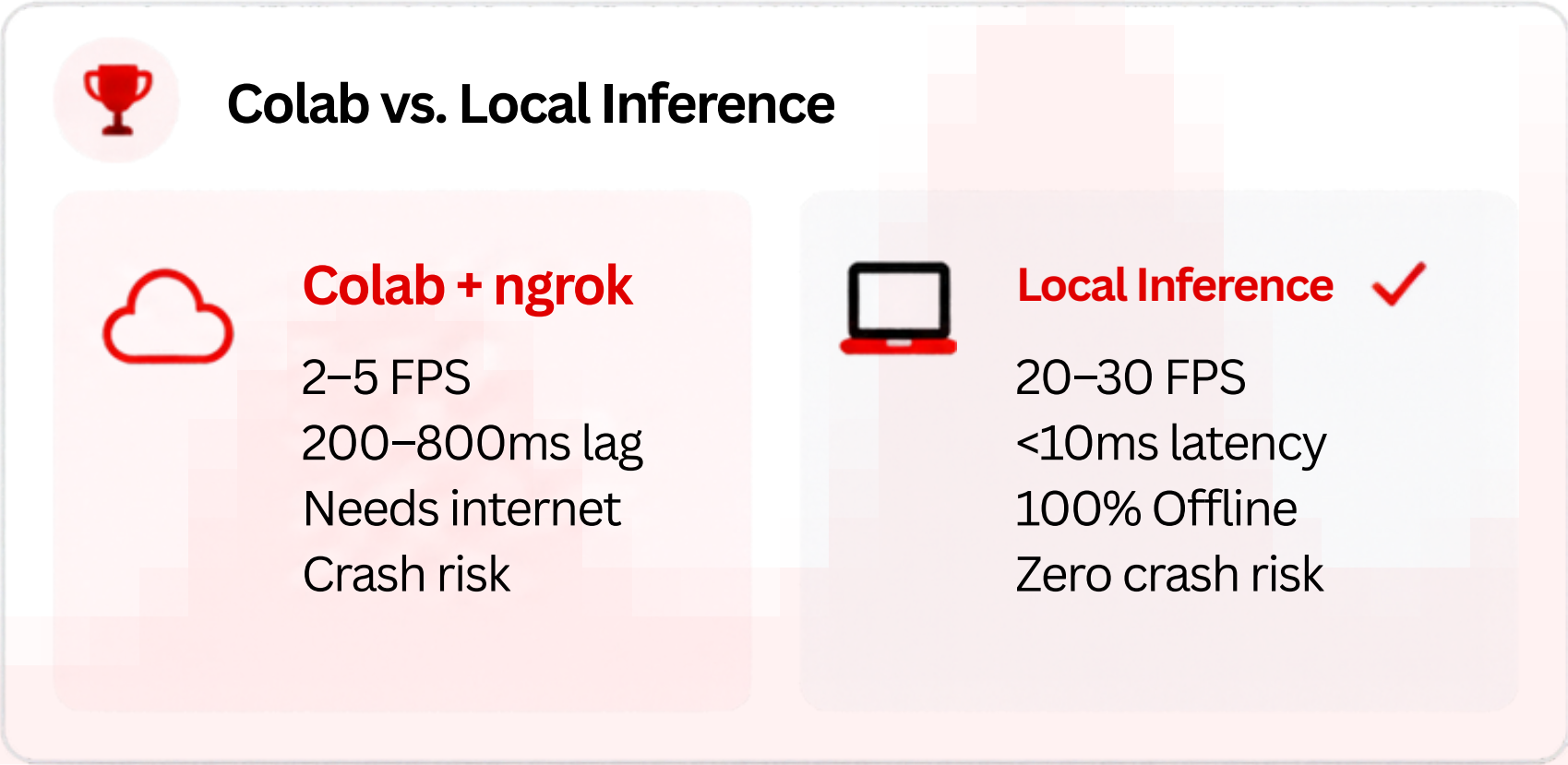
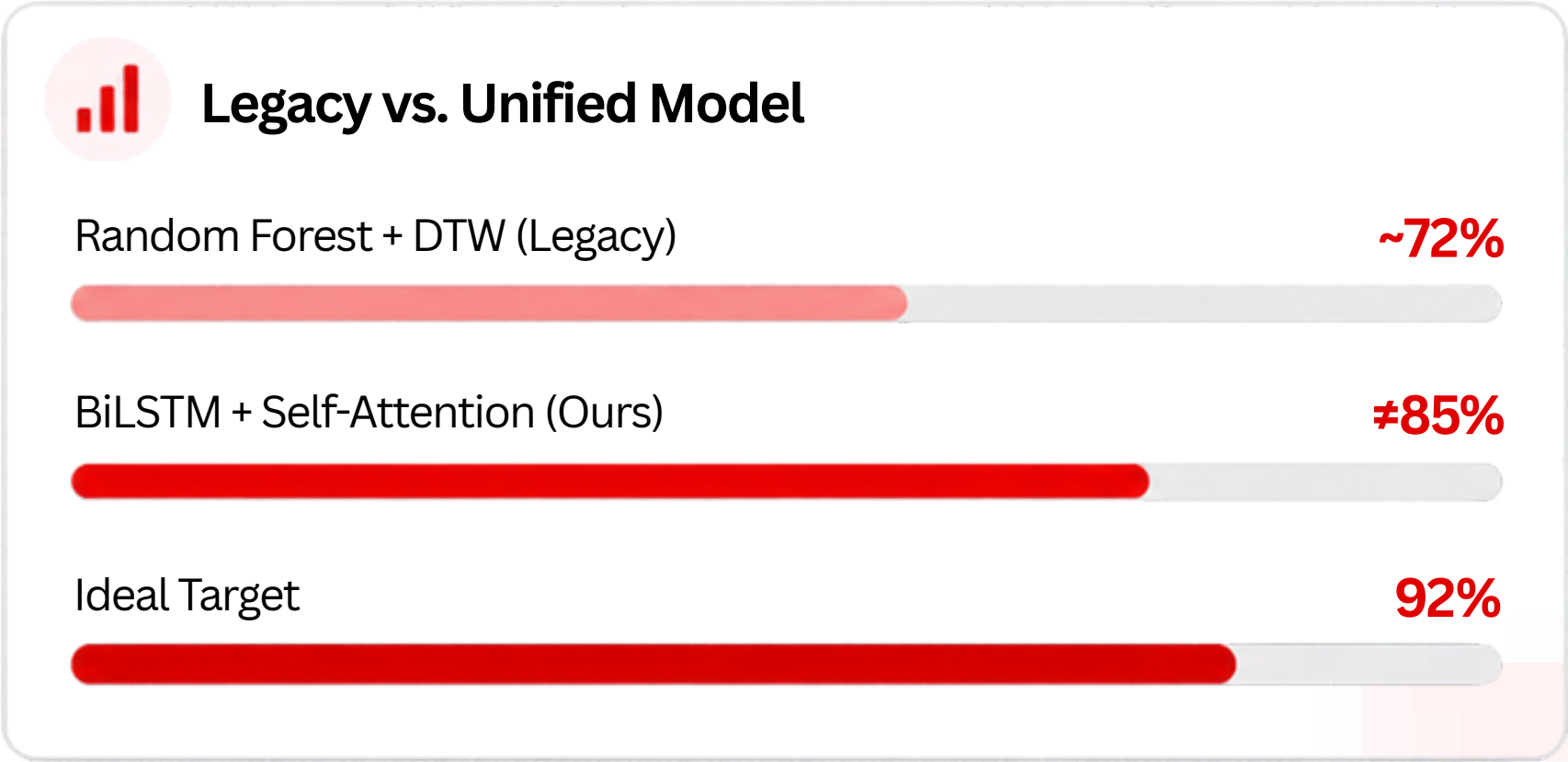


## Tab 3 — Architecture Info

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Model specs, input dimensions, inference latency, and training details

# Target Metrics & Performance Benchmarks



# Thank you

